

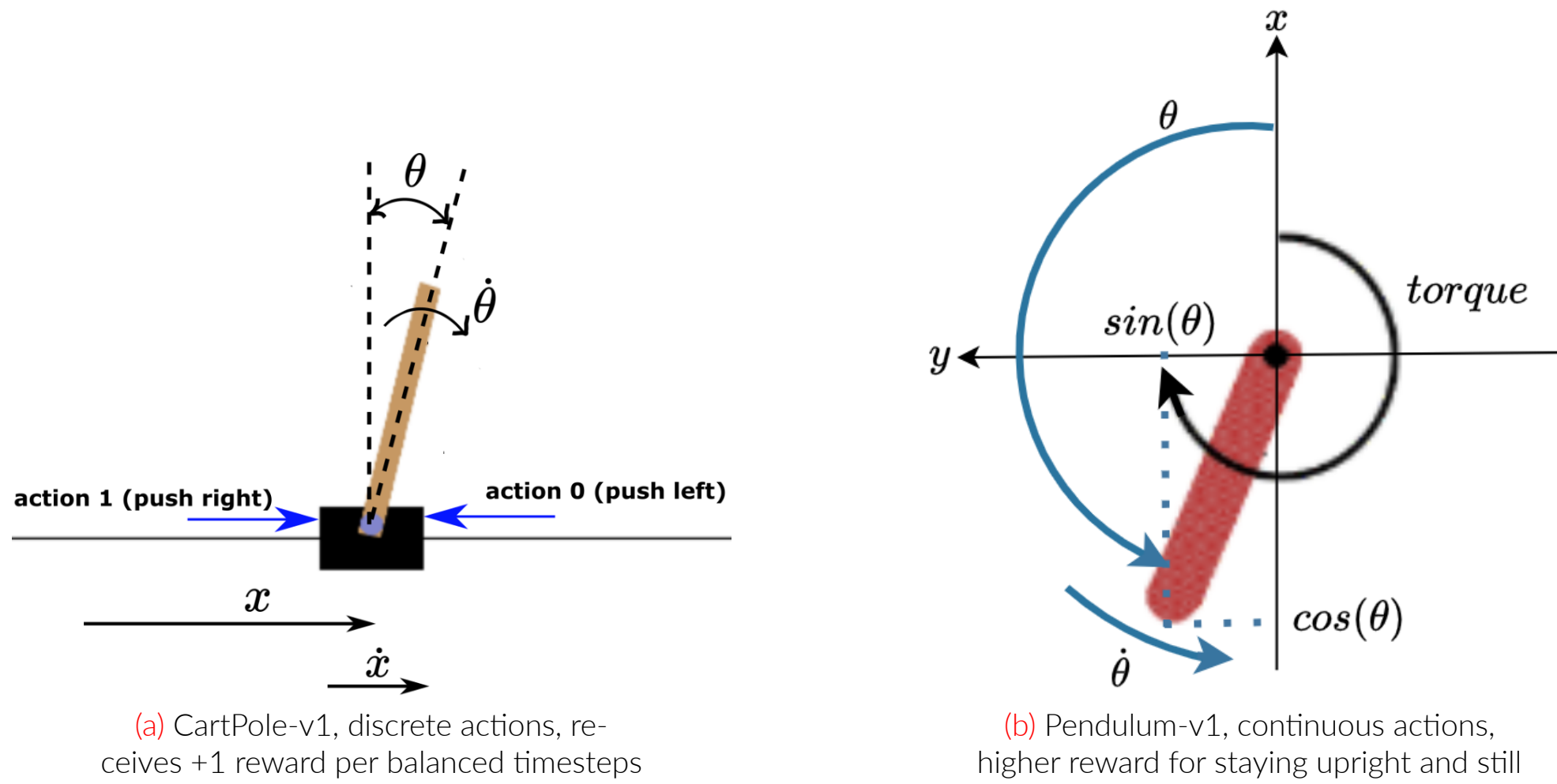
Return of the Policy: Applied Project 3

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Context

- **Imitation Learning (IL)** seeks to train agents to perform tasks by mimicking expert demonstrations, reducing the need for environment rewards.
- We evaluate IL algorithms on classic OpenAI-Gym environments: **CartPole-v1** (fig. 2e) and **Pendulum-v1** (fig. 2f).
- Expert demonstrations are generated by training RL agents using the **Proximal Point Optimization (PPO)** algorithm.
- We compare the performance of three imitation learning approaches : **IQ-Learn**, **F-IRL** and the **SOAR version of F-IRL**, analyzing sample efficiency and robustness across different demonstration dataset sizes.
- All results are averaged over 3 seeds for statistical significance.



Training experts with PPO

- We used PPO because it is simple to implement compared to TRPO, and works well empirically. It relies in clipping the distance between policies in the loss function.
- **Innovations** : We use an exponential learning rate decay to avoid too fast parameter tuning in later stages of training.

IQ-Learn

- This method learns a Q-function from demonstrations without requiring access to rewards or adversarial training. The key advantages is the stability of the method, since it does not require an adversary.
- **Innovations** : we used a discriminator neural network to distinguish between expert and agent. The output of this discriminator are Q-values, and are trated as rewards. They are then shaped using this discriminator, which assign high probability to expert actions and penalize overconfidence in agent actions. Finally we use PPO to update policy and value network, based on this Q-value.

f-IRL

- **f-divergence Inverse (Imitation) Reinforcement Learning (f-IRL)** learns rewards by matching expert and policy feature expectation, by minimizing the error between the mean feature vector from expert trajectory and from current trajectory.
- **Implementation** : The f-divergence was implemented using the Kullback-Leibler (KL) divergence, the densities are estimated using Kernel Density Estimation (KDE). The reward and policy updated are performed iteratively, the reward updated using f-divergence and the policy is then updated using that reward and PPO.[3]
- We chose both continuous and a discrete environment, so we chose to use F-IRL, because it's able to work with both types of environments.

SOAR

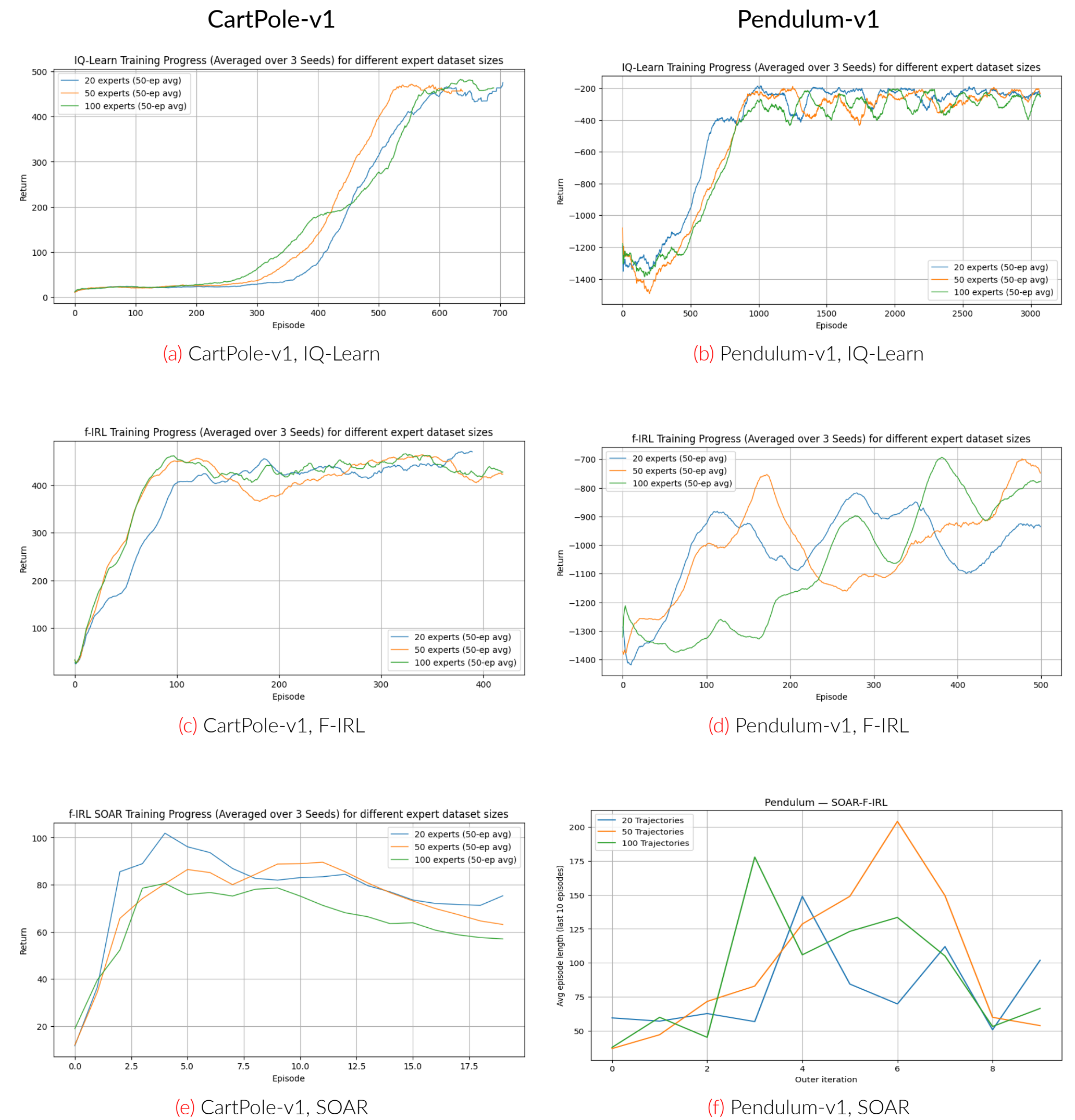
- **Soft Optimistic Actor-cRitic (SOAR)** improves imitation learning by using an optimistic critic within SAC that underestimates the expected cumulative cost. This optimism encourages better exploration under uncertainty, leading to more efficient policy updates.[2][5]
- **Innovations** : SOAR improves exploration in imitation learning by using an ensemble of critics to compute optimistic value estimates, replacing the single critic in Soft Actor Critic. This encourages better state-action coverage and scales efficiently to deep neural networks in continuous control tasks.

References

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Results

The algorithms were tested on the two environments using different sized datasets, 20, 50 and 100 expert trajectories. For each of these datasets the performance of the algorithms was averaged over 3 seeds.



Observations

IQ-Learn

- **CartPole**: Initially, training reward stagnated, but improvement correlated with expert dataset size, 100 experts led to faster learning than 50 or 20. Eventually, performance plateaued for all datasets around a reward of 450 (out of 500). This suggests IQ-Learn is effective even with few expert demonstrations in simple, discrete environments. To ensure high-quality training data, only PPO expert trajectories achieving the maximum reward (500) were used.
- **Pendulum**: IQ-Learn improves steadily on Pendulum, reaching rewards between -400 and -200. Smaller expert datasets learn faster initially but plateau lower, while larger ones converge more effectively in the long run. Results also highlight the importance of expert quality, only using demonstrations above -200 yielded more meaningful training.

f-IRL

- **CartPole**: f-IRL performs well on the CartPole environment, generating effective rewards and policies. Larger expert datasets lead to faster and more stable learning, with the 100-expert set producing the most consistent returns, reaching around 450.
- **Pendulum**: f-IRL struggles on the Pendulum environment, with final returns between -950 and -750, indicating it fails to solve the task. Performance varies greatly across seeds, even with the same dataset, highlighting the algorithm's sensitivity to expert quality and high variance in outcomes. No consistent pattern was observed regarding dataset size.

SOAR

- **CartPole**: SOAR-enhanced f-IRL demonstrated poor performance on the CartPole environment, achieving rewards significantly below both IQ-Learn and standard f-IRL. The algorithm struggled to learn effectively, showing high sensitivity to the specific expert trajectories and hyperparameters. It also has a large computational overhead, making training considerably more expensive compared to other methods.
- **Pendulum**: SOAR-enhanced f-IRL performed poorly on the Pendulum task, mainly due to low-quality expert trajectories (since f-IRL also struggled to solve the task). Even with extensive hyperparameter tuning, it failed to learn effectively and remained computationally expensive.

Conclusion

Our experiments demonstrated clear distinctions between imitation learning methods in terms of robustness, sample efficiency, and sensitivity to demonstration dataset size.

IQ-Learn consistently achieved stable learning across both discrete (CartPole-v1) and continuous (Pendulum-v1) environments. Notably, dataset size significantly impacted the convergence speed for CartPole but had minimal influence on Pendulum, suggesting that Pendulum's competitive dynamics might reduce the dependence on extensive expert demonstrations.

F-IRL showed robust performance on CartPole-v1 but performed poorly on Pendulum-v1. The method exhibited high variance and was significantly dependent on the specific expert trajectories, limiting its reliability in the continuous environment. However, varying the expert dataset size did not notably impact convergence speed for CartPole, indicating some resilience to limited demonstrations in discrete settings.

SOAR-augmented F-IRL faced significant challenges across both environments, with performance falling short of expectations despite the theoretical promise of optimistic exploration. The method exhibited notable sensitivity to hyperparameter configuration and expert demonstration quality, while requiring substantially greater computational resources due to its multi-critic architecture. The high computational cost limited extensive hyperparameter exploration, suggesting that further refinements may be needed to achieve satisfactory performance in practical applications.